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Paper Id: 231898 Roll No. | | | | | | | | | | |

B.PHARM

(SEM VII) THEORY EXAMINATION 2022-23

PHARMACEUTICS-IX (BIOPHARMACEUTICS & PHARMACOKINETICS)

Time: 3 Hours ***Total Marks: 70***

Note: Attempt all Sections.

SECTION A

1. Attempt *all* questions in brief. 2 x 7 = 14

(a) What is Facilitated diffusion?

(b) Mention the limitations of one compartment model.

(c) Enlist objectives of bioavailability studies.

(d) What does the word 'open' mean in two-compartment model?

(e) Why nonlinear kinetics are called dose-dependent kinetic?

(f) Define Cmax and AUC.

(g) Define Zero order and First order absorption rate constant.

SECTION B

2. Attempt any *three* of the following: 7 x 3 = 21

(a) Detail the various theories for the dissolution of the drug.

(b) What is a compartment? Classify the compartment models. Give the schematic representation of the same

(c) Define IVIVC. Discuss its significance.

(d) Calculate the various parameters and Ka with the help of the urine data method.

(e) Discuss the concept of drug clearance.

SECTION C

3. Attempt any *one* part of the following: 7 x 1 = 7

(a) Explain the concept of determination of Ka by the Loo Ringelmann method.

(b) What is the significance of plasma drug concentration in the calculation of various pharmacokinetic parameters?

4. Attempt any *one* part of the following: 7 x 1 = 7

(a) Discuss in detail one-compartment open model for a drug administered as an I.V. infusion

(b) Discuss in detail the two-compartment open model for a drug administered as IV Bolus.

5. Attempt any *one* part of the following: 7 x 1 = 7

(a) Define clinical pharmacokinetics. Explain its scope in detail.

(b) Discuss the various methods of dosage adjustment in renal failure patient.

6. Attempt any *one* part of the following: 7 x 1 = 7

(a) Define absorption. Explain the various mechanism of drug transport.

(b) Discuss the patient-related factors affecting drug absorption.

7. Attempt any *one* part of the following: 7 x 1 = 7

(a) What are the various methods of enhancement of bioavailability?

(b) Define bio-equivalence. List the various methods involved in its determination

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